Exercise 8.1. *Prove that if $f \in \mathcal{C}^\infty$*

$$\text{上确界} \quad \sup_{k\in\mathbb{Z}} |k|^m |\tilde{f}(k)| < \infty, \quad \forall m \in \mathbb{N}.$$

Hint: Use Schwarz inequality, and you may first consider $m = 1$.

$m=1$ $\quad |k||\tilde{f}(k)| = |k| \left| \frac{1}{2\ell} \int_{-\ell}^{\ell} f(x) e^{-i\frac{k\pi x}{\ell}} dx \right|$

$$= \left| \frac{k}{2\ell} \int_{-\ell}^{\ell} f(x) e^{-i\frac{k\pi x}{\ell}} dx \right| < \left|\frac{k}{2\ell}\right| \int_{-\ell}^{\ell} |f(x)| dx$$

if $f \in \mathcal{C}^\infty$ $\quad \exists x_0 \in [-\ell, \ell]$, $f(x) \leq f(x_0) < \infty$

$$\Rightarrow \quad |k||\tilde{f}(k)| < \left|\frac{k}{2\ell} 2\ell f(x_0)\right| = |k f(x_0)| < \infty$$

$m>1$ $\quad |k|^m |\tilde{f}(k)| = |k|^m \left| \frac{1}{2\ell} \int_{-\ell}^{\ell} f(x) e^{-i\frac{k\pi x}{\ell}} dx \right|$

$$= |k|^m \left| \frac{1}{2\ell} \int_{-\ell}^{\ell} f(x) \frac{\ell}{ik\pi} d e^{-i\frac{k\pi x}{\ell}} \right|$$

$$= |k|^m \left| \frac{1}{2\ell} \left(\frac{\ell}{ik\pi} f(x) e^{-i\frac{k\pi x}{\ell}} \Big|_{-\ell}^{\ell} - \int_{-\ell}^{\ell} f^{(1)}(x) \frac{\ell}{ik\pi} e^{-i\frac{k\pi x}{\ell}} dx \right) \right|$$

$$= |k|^m \left| \frac{1}{2\ell} \left(\frac{\ell}{ik\pi} e^{ik\pi} [f(\ell) - f(-\ell)] - \frac{\ell}{ik\pi} \int_{-\ell}^{\ell} f^{(1)}(x) e^{-i\frac{k\pi x}{\ell}} dx \right) \right|$$

$$= |k|^m \frac{1}{2\ell} \left| \sum_{n=1}^{m} \left(\frac{\ell}{ik\pi}\right)^n (-1)^{n+1} e^{ik\pi} [f^{(n)}(\ell) - f^{(n)}(-\ell)] + (-1)^m \left(\frac{\ell}{ik\pi}\right)^m \int_{-\ell}^{\ell} f^{(m)}(x) e^{-i\frac{k\pi x}{\ell}} dx \right|$$

$$= \frac{1}{2\ell} \left| \sum_{n=1}^{m} k^{m-n} \left(\frac{\ell}{i\pi}\right)^n e^{i(n+k+1)\pi} [f^{(n)}(\ell) - f^{(n)}(-\ell)] + (-1)^m \left(\frac{\ell}{i\pi}\right)^m \int_{-\ell}^{\ell} f^{(m)}(x) e^{-i\frac{k\pi x}{\ell}} dx \right|$$

$T = 2\ell \quad f^{(n)}(\ell) - f^{(n)}(-\ell) = 0$?

$$\Rightarrow \quad |k|^m |\tilde{f}(k)| = \frac{1}{2\ell} \left| \left(\frac{\ell}{i\pi}\right)^m \int_{-\ell}^{\ell} f^{(m)}(x) e^{-i\frac{k\pi x}{\ell}} dx \right| \leq C_m = \text{const}$$

Exercise 8.5. *Prove that the orthonormality of the irreducible characters (8.2) implies the orthonormaltiy of* $\{\cos\left(\frac{k\pi x}{l}\right), \sin\left(\frac{k\pi x}{l}\right) | k = 0, 1, 2, \dots\}$.

$$\chi^k(x) = e^{i\frac{k\pi x}{\ell}}, \quad x \in [-\ell, \ell] \qquad \int_{-\ell}^{\ell} e^{i\frac{j\pi x}{\ell}} e^{-i\frac{k\pi x}{\ell}} dx = 2\ell\,\delta_{jk}$$

$$\int_{-\ell}^{\ell} \sin\frac{j\pi x}{\ell} \cos\frac{k\pi x}{\ell}\, dx = \int_{-\ell}^{\ell} \cos\frac{j\pi x}{\ell} \sin\frac{k\pi x}{\ell}\, dx = 0 \qquad \text{odd func}$$

$j, k \neq 0$ 时

$$\int_{-\ell}^{\ell} \cos\frac{j\pi x}{\ell} \cos\frac{k\pi x}{\ell}\, dx = \frac{1}{2}\int_{-\ell}^{\ell} \cos\frac{j+k}{\ell}\pi x + \cos\frac{j-k}{\ell}\pi x\, dx$$

$$= \frac{1}{2}\mathrm{Re}\int_{-\ell}^{\ell} e^{i\frac{j+k}{\ell}\pi x} + e^{i\frac{j-k}{\ell}\pi x}\, dx$$

$$= \frac{1}{2}\mathrm{Re}\,(2\ell\,\delta_{j,-k} + 2\ell\,\delta_{j,k})$$

$$= \ell\,\delta_{jk} \qquad \text{since } j, k \in N$$

Exercise 8.7. *Fourier-series expand* $\frac{\sin x}{x}$ *in the interval* $(0, \pi)$. *From the coefficients of the expansion you can compute*

$$\int_0^\infty \frac{\sin x}{x} \mathrm{d}x.$$

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{1}{2}\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \frac{1}{2}\left(\mathrm{Im}\int_{-\infty}^{\infty} \frac{e^{iz}}{z} dz\right)$$

$$= \frac{1}{2}\mathrm{Im}\left(-\int_{C_\infty} \frac{e^{iz}}{z} dz - \underline{\int_{C_\delta} \frac{e^{iz}}{z} dz}\right) \qquad \int_0^{-\pi} \frac{e^{i\delta}}{z}\, iz\, d\theta = -i\pi$$

$$= \frac{\pi}{2}$$

Exercise 8.11. *Try to derive the Fourier transform (8.51) and inverse Fourier transform 8.51 from Fourier series in the limit $2l \to \infty$.*

$$T = 2l, \quad f(x) = \sum_{k \in \mathbb{Z}} \tilde{f}(k) e^{i\frac{k\pi x}{l}} \qquad \tilde{f}(k) = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-i\frac{k\pi x}{l}} dx$$

$$\omega \equiv \frac{k\pi}{l}$$

$$f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \tilde{f}(\omega) e^{i\omega x} d\omega$$

$$\tilde{f}(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

Exercise 8.13. *Given that $xf \in L^2(\mathbb{R})$ and $f' \in L^2(\mathbb{R})$, prove that $f \in L^2(\mathbb{R})$. Hint: Use the Cauchy-Schwarz inequality.*

$$\int_{\mathbb{R}} f f' dx = \int_{\mathbb{R}} f df = \frac{1}{2} f^2 \Big|_{-\infty}^{\infty} - \int_{\mathbb{R}} f' f dx$$

$$\int_{\mathbb{R}} |f|^2 dx = \int_{\mathbb{R}} \frac{f}{x} \cdot xf \, dx \leq \sqrt{\int_{\mathbb{R}} \left|\frac{f}{x}\right|^2 dx} \sqrt{\int_{\mathbb{R}} |xf|^2 dx}$$

Exercise 8.15. *One can define the deviation or dispersion of $f(x)$ about any point $x = a$ as*

$$\Delta_a = \int_{\mathbb{R}} (x-a)^2 |f(x)|^2 \mathrm{d}x.$$

Prove that

$$\Delta_a f \Delta_b \tilde{f} \geq \frac{1}{4}. \tag{8.64}$$

Exercise 8.17. *If $f(x)$ is real or purely imaginary, what conditions on $\tilde{f}(\omega)$ does the above relation impose?*

$f(x)$ is real: $\tilde{f}^*(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx = \tilde{f}(\omega)$ $\tilde{f}(\omega)$ is real

$f(x)$ is imag: $\tilde{f}^*(\omega) = -\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx = -\tilde{f}(\omega)$ $\tilde{f}(\omega)$ is imag

Exercise 8.19. *Given the definition of Fourier transform in $\mathbb{R}^n$ in Eqs. (8.54) and (8.55), generalize the above properties of Fourier transform to n dimensions.*

$$\tilde{f}(\vec{k}) = \frac{1}{\sqrt[n]{2\pi}} \int_{\mathbb{R}^n} f(\vec{x}) \mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{x}} \mathrm{d}^n\vec{x},$$

$\in L(\mathbb{R}^n)$ too, the inverse Fourier tra

$$f(\vec{x}) = \frac{1}{\sqrt[n]{2\pi}} \int_{\mathbb{R}^n} \tilde{f}(\vec{k}) \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}} \mathrm{d}^n\vec{k},$$

Linearity: any 2 integrable funcs $f(x)$ $g(x)$ and $a, b \in \mathbb{C}$

$$\widetilde{af + bg} = a\tilde{f} + b\tilde{g}$$

Translation: for $f(\vec{x})$ $g(\vec{x}) = f(\vec{x} - \vec{x}_0)$

$$\tilde{g}(\vec{k}) = \frac{1}{\sqrt[n]{2\pi}} \int_{\mathbb{R}^n} f(\vec{x} - \vec{x}_0) e^{-i\vec{k}\cdot\vec{x}} d^n\vec{x}$$

仿 8.54 & 8.55 写即可.

Exercise 8.21. *The Bessel functions $J_n(x)$ may be defined from the Fourier-series*

$$e^{ix\sin t} = \sum_{n=-\infty}^{\infty} J_n(x) e^{int}$$

Assume $x \in \mathbb{R}$, compute the infinite sum

$$\sum_{n=-\infty}^{\infty} |J_n(x)|^2 .$$

Parseval's thm: $A(x) = \sum_{n=-\infty}^{\infty} a_n e^{inx}$ $\quad B(x) = \sum_{n=-\infty}^{\infty} b_n e^{inx}$

$$\sum_{n=-\infty}^{\infty} a_n \overline{b_n} = \frac{1}{2\pi} \int_{-\infty}^{\infty} A(x) \overline{B(x)} \, dx$$

$$\sum_{n=-\infty}^{\infty} |J_n(x)|^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ix\sin t} e^{-ix\sin t} \, dx = 1$$

Exercise 8.23. *Compute the following integrals using Fourier transform.*

1.
$$\int_{-\infty}^{\infty} \frac{\sin(ax)\sin(bx)}{x^2} dx, \quad a, b \geq 0,$$

2.
$$\int_{-\infty}^{\infty} \frac{\sin x}{x\,(x^2+1)} dx.$$

1. $\int_{-\infty}^{\infty} \frac{\sin ax}{x} e^{ikx} \, dx = \begin{cases} \pi & |k|<a \\ 0 & |k|>a \end{cases}$

Parseval's thm $\int_{-\infty}^{\infty} dx \, f(x)\overline{g}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \, \tilde{f}(k) \overline{\tilde{g}}(k)$

$$\int_{-\infty}^{\infty} dx \, \frac{\sin ax}{x} \frac{\sin bx}{x} = \frac{\pi^2}{2\pi} 2 \min\{a,b\} = \pi \min\{a,b\}$$

2. $\int_{-\infty}^{\infty} \frac{1}{x^2+1} e^{ikx} \, dx = e^{-|k|} \pi$

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \frac{1}{x^2+1} dx = \frac{1}{2\pi} \pi \pi \, (e^{-1} + 1 + e^{-1}) = \pi(\tfrac{1}{2} + \tfrac{1}{e})$$

? $\pi(1-\frac{1}{e})$ mma 结果

Exercise 8.25. *Adapt the properties of complex-valued Fourier transform to Fourier sine and cosine transform.*

参考 Linearity ...

Exercise 8.27. *Prove that the functions* $\varphi_n(x) = \frac{\sin\frac{x}{2}}{\pi x}\mathrm{e}^{\mathrm{i}nx}$, $n \in \mathbb{Z}$, *are square-integrable and pairwise orthogonal. Find the numbers* a_n *that minimizes*

$$\int_{-\infty}^{\infty}\left|\frac{1}{1+x^2} - \sum_{n=-N}^{N} a_n\varphi_n(x)\right|^2 \mathrm{d}x.$$

Is $\{\varphi_n | n \in \mathbb{Z}\}$ *a complete basis of* $L^2(\mathbb{R})$*?* yes ?

$$\int_{-\infty}^{\infty} \varphi_n(x)\varphi_n^*(x)\,dx = \int_{-\infty}^{\infty} \frac{\sin\frac{x}{2}}{\pi x}\frac{\sin\frac{x}{2}}{\pi x}\,dx = \frac{1}{2\pi}$$

$$\int_{-\infty}^{\infty} \varphi_n(x)\varphi_m^*(x)\,dx = \int_{-\infty}^{\infty} \frac{\sin\frac{x}{2}}{\pi x}\frac{\sin\frac{x}{2}}{\pi x}e^{i(n-m)x}\,dx = \frac{1}{2\pi}\delta_{mn}$$

$$\frac{1}{1+x^2} = \sum_{m=-N}^{N} a_m\varphi_m(x) \qquad \int_{-\infty}^{\infty} dx\,\frac{1}{1+x^2}\varphi_n^*(x) = \int_{-\infty}^{\infty}\sum_{m=-N}^{N} a_m\varphi_m(x)\varphi_n^*(x)\,dx$$

$$= \frac{1}{2\pi}a_n$$

$$\begin{cases} a_n = \frac{\pi}{\sqrt{e}}\left(e^{n+1} - 2\sinh(n)\right) & \text{mma积的} \\ n+n^* \le -1 \text{ 时} \end{cases}$$

Exercise 8.29. *Let $H(x,y), (x,y) \in \mathbb{R}^2$, be the Heaviside function on $\mathbb{R}^2$, namely:*

$$H(x) = \begin{cases} 1, & x \geq 0, y \geq 0 \\ 0, & otherwise \end{cases}$$

Prove that

$$\frac{\partial^2 H(x,y)}{\partial x \partial y} = \delta(x,y).$$

Generalize the above to the derivative of the Heaviside function on $\mathbb{R}^n$.

对 $\forall f(x,y) \in L^2$ $\int_{\mathbb{R}^2} f(x,y) \frac{\partial^2 H(x,y)}{\partial x \partial y} dx dy = \int_{\mathbb{R}^2} f(x,y)\, dx\, d\frac{\partial H(x,y)}{\partial x}$

$$= -\int_{\mathbb{R}^2} \frac{\partial H(x,y)}{\partial x} \frac{\partial f(x,y)}{\partial y} dx\, dy$$

$$= -\int_{\mathbb{R}^2} \frac{\partial f(x,y)}{\partial y} dH(x,y)\, dy$$

$$= \int_{\mathbb{R}^2} H(x,y) \frac{\partial^2 f(x,y)}{\partial x \partial y} dx\, dy$$

$$= \int_0^\infty \int_0^\infty \frac{\partial^2 f(x,y)}{\partial x \partial y} dx\, dy$$

$$= \int_0^\infty \frac{\partial f(x,y)}{\partial x}\Big|_0^\infty dx = -\int_0^\infty \frac{\partial f(x,y=0)}{\partial x} dx$$

$$= \int_{\mathbb{R}^2} f(x,y)\, \delta(x,y)\, dx\, dy$$ 证毕.

Exercise 8.31. *If $F(x) = 0$ has nonsimple roots, say, double roots or triple roots, how is Eq. 8.104 going to be rewritten?*

Poisson 求和公式

Exercise 8.33. *Consider the function $f(r) = \delta(r-a)/r$, where $r = \sqrt{x^2+y^2+z^2}$ and $a \in \mathbb{R}$. Fourier-transform $f(r)$. You may use spherical coordinates.*

$$\frac{1}{\sqrt[3]{2\pi}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{e^{-i\vec{k}\cdot\vec{r}}\,\delta(r-a)}{r}\,dx\,dy\,dz = \frac{1}{\sqrt[3]{2\pi}}\int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} \frac{e^{-ikr\cos\theta}}{r}\,\delta(r-a)\,r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= \frac{2\pi a}{\sqrt[3]{2\pi}}\underbrace{\int_0^{\pi} e^{-iak\cos\theta}\sin\theta\,d\theta}_{\frac{2\sin(ak)}{ak}}$$

$$= \frac{4\pi}{\sqrt[3]{2\pi}}\,\frac{\sin ak}{k} \equiv \tilde{f}(k)$$

Exercise 8.35. *Use the definition of the differential operator $\mathcal{D}_\alpha$ in Eq. (8.96) to generalize Eq. (8.110) to n dimensions.*

$(\partial_\alpha T)(\varphi) = (-1)^{|\alpha|}\,T(\partial_\alpha\varphi)$ 哪个 $\mathcal{D}_\alpha$?

Exercise 8.37. *Evaluate the RHS of (8.115) to see if you would obtain the same result as Eq. (8.113).*

(8.115) $I(r) = \int_{\mathbb{R}} \lim_{\epsilon\to 0} f_\epsilon(p)\,e^{ipr}\,dp = \lim_{\epsilon\to 0}\int_{\mathbb{R}} f_\epsilon(p)\,e^{ipr}\,dp$

Exercise 8.39. *Find the Fourier transform of the Heaviside step function (8.94). Hint: Use the ϵ-prescription.*

$$H(x-a)=\begin{cases}1 & x\geq a\\ 0 & x<a\end{cases}$$

$$\tilde{H}(k)=\int_{-\infty}^{\infty}H(x-a)e^{-ikx}dx=\int_{-\infty}^{\infty}H(x)e^{-ik(x+a)}dx$$

$$=\lim_{\epsilon\to 0}\int_0^{\infty}e^{-\epsilon x}e^{-ikx}e^{-ika}dx$$

$$=\lim_{\epsilon\to 0}(\epsilon+ik)^{-1}e^{-ika}$$

$$=e^{-ika}\lim_{\epsilon\to 0}\frac{1}{\epsilon+\frac{k^2}{\epsilon}}-\frac{ik}{\epsilon^2+k^2}$$

$$=e^{-ika}\left[\pi\delta(k)-P\frac{i}{k}\right]$$

P for Cauchy's principal value

Exercise 8.41. *The partition function of a single harmonic oscillator is*

$$z(\beta)=\sinh\left(\frac{\beta\hbar\omega}{2}\right)^{-1}.$$

Find the density of states $w(E)$.

$$z(\beta)=\frac{2}{e^{\beta\hbar\omega/2}-e^{-\beta\hbar\omega/2}}=2\frac{e^{-\hbar\omega\beta/2}}{1-e^{-\hbar\omega\beta}}\qquad Z_N=Z_1^N=2^Ne^{-\frac{N\hbar\omega\beta}{2}}(1-e^{-\hbar\omega\beta})^{-N}$$

$$w(E)=\frac{2^N}{2\pi i}\int_{\beta'-i\infty}^{\beta'+i\infty}d\beta\, e^{\beta(E-\frac{\hbar\omega}{2})}(1-e^{-\hbar\omega\beta})^{-N}$$

$$=\frac{2^N}{2\pi i}\int_{\beta'-i\infty}^{\beta'+i\infty}e^{\beta E}d\beta\, e^{-\frac{\beta\hbar\omega}{2}}(1-e^{-\hbar\omega\beta})^{-N}$$

$$=\frac{2^N}{2\pi i}\int_{\beta'-i\infty}^{\beta'+i\infty}e^{\beta E}\sum_{M=0}^{\infty}C_{-N}^{M}(-1)^Me^{-(M+\frac{1}{2})\hbar\omega\beta}d\beta$$

$$=\frac{2^N}{2\pi i}\int_{\beta'-i\infty}^{\beta'+i\infty}e^{\beta E}\sum_{M=0}^{\infty}\frac{(N+M-1)!}{(N-1)!\,M!}e^{-(M+\frac{1}{2})\hbar\omega\beta}d\beta$$

$$=\sum_{M=0}^{\infty}\frac{(N+M-1)!}{(N-1)!\,M!}\frac{2^N}{2\pi i}\int_{\beta'-i\infty}^{\beta'+i\infty}e^{\beta[E-(M+\frac{1}{2}N)\hbar\omega]}d\beta$$

$$=2^N\sum_{M=0}^{\infty}\frac{(N+M-1)!}{(N-1)!\,M!}\delta[E-(M+\tfrac{1}{2})\hbar\omega]$$

where $\frac{1}{2\pi i}\int_{\beta'-i\infty}^{\beta'+i\infty}e^{a\beta}d\beta=\delta(a)$

即 number of arrangement for $E=(M+\frac{1}{2})\hbar\omega$ is $2^N\frac{(N+M-1)!}{(N-1)!\,M!}$ 可分别

- **Translation.**

$$f(t-t_0) \fallingdotseq e^{-pt_0}\bar{f}(p). \tag{8.123}$$

Exercise 8.43. *Prove Eq.* (8.123).

$$\bar{f}(p) = \int_0^\infty e^{-pt} f(t)\,dt$$

$$\int_{-t_0}^\infty f(t-t_0)e^{-pt}\,dt = \int_0^\infty f(t)e^{-p(t+t_0)}\,dt = e^{-pt_0}\bar{f}(p)$$

Exercise 8.45. *Prove that in Fig.* 8.7,

$$\lim_{\epsilon\to 0}\int_{C_\epsilon}\bar{f}(p)e^{pt}\,dp = 0.$$

$$\bar{f}(p) = \frac{1}{\sqrt{p}} \qquad p = \rho e^{i\theta} \qquad \lim_{\epsilon\to 0}\int_{C_\epsilon}\bar{f}(p)e^{pt}\,dp = \lim_{\rho\to 0}\left(\frac{1}{\sqrt{p}}e^{pt}\,i\rho\right) = 0$$

Exercise 8.47. *Solve the equation of an alternating RC circuit*

$$RJ(t) + \frac{1}{C}\int_0^t J(t')\mathrm{d}t' = E_0\sin(\omega t),$$

with initial condition $J(0) = 0$.

两边微分 $R\frac{dJ}{dt} + \frac{1}{C}J(t) = E_0\omega\cos(\omega t)$

齐次解 $R\frac{dJ}{dt} + \frac{1}{C}J = 0 \qquad J_h(t) = Ae^{-\frac{t}{RC}}$

非齐 $J_p(t) = B\cos(\omega t) + C\sin(\omega t)$ 代回

$$R(-B\omega\sin(\omega t) + C\omega\cos(\omega t)) + \frac{1}{C}(B\cos(\omega t) + C\sin(\omega t)) = E_0\omega\cos(\omega t)$$

$$\begin{cases} -RB\omega + 1 = 0 \\ RC\omega + \frac{B}{C} = E_0\omega \end{cases} \qquad B = \frac{1}{R\omega} \qquad C = \frac{1}{2}\left[\frac{E_0}{R} \pm \sqrt{\left(\frac{E_0}{R}\right)^2 - \frac{4}{(R\omega)^2}}\right]$$

$$J(t) = J_h(t) + J_p(t) = Ae^{-\frac{t}{RC}} + B\cos(\omega t) + C\sin\omega t$$

$$J(0) = 0 \Rightarrow A = -B = -\frac{1}{R\omega}$$

Exercise 8.49. *By means of Laplace transform, evaluate the two integrals*

$$\int_0^\infty \frac{\cos(tx)}{x^2+a^2}\mathrm{d}x \qquad \text{and} \qquad \int_0^\infty \frac{\sin(tx)}{x(x^2+1)}\mathrm{d}x.$$

$$L[e^{-ax}] = \int_0^\infty e^{-(a+p)x}\,dx = \frac{1}{a+p} \qquad L^{-}\left[\frac{1}{a+p}\right] = e^{-ax}$$

$$L[1-e^{-ax}] = \int_0^\infty e^{-px} - e^{-(a+p)x}\,dx = \frac{a}{p(a+p)} \qquad L^{-}\left[\frac{1}{p(1+p)}\right] = 1-e^{-x}$$

令 $I_1(t) = \int_0^\infty \frac{\cos(tx)}{x^2+a^2}\,dx$ $\qquad F(p) = L[I_1(t)] = \int_0^\infty\int_0^\infty \frac{\cos tx}{x^2+a^2} e^{-pt}\,dx\,dt$

$$= \int_0^\infty \cos tx\, e^{-pt}\,dt \int_0^\infty \frac{1}{a^2+x^2}\,dx$$

$$= \int_0^\infty \frac{p}{p^2+x^2}\,\frac{1}{a^2+x^2}\,dx = \frac{\pi}{2a}\,\frac{1}{a+p}$$

两边作逆变换 $\quad I_1(t) = \frac{\pi}{2a} e^{-at}$

$I_2(t) = \int_0^\infty \frac{\sin(tx)}{x(x^2+1)}\,dx$ $\qquad F(P) = L[I_2(t)] = \int_0^\infty\int_0^\infty \frac{\sin tx}{x(x^2+1)} e^{-pt}\,dx\,dt$

$$= \int_0^\infty \sin tx\, e^{-pt}\,dt \int_0^\infty \frac{1}{x(x^2+1)}\,dx$$

$$= \int_0^\infty \frac{x}{p^2+x^2}\,\frac{1}{x(x^2+1)}\,dx = \frac{\pi}{2p}\,\frac{1}{1+p}$$

两边作逆变换 $\quad I_2(t) = \frac{\pi}{2}(1-e^{-t})$